IIT KANPUR

ME772A: MECHATRONICS

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Application Of Swarm Robotics To Combat Oil Spillage In Water Bodies

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Abstract:

People across the globe are trying to take precautions for the problem of water pollution. One of the major causes of water pollution is the Oil Spills. Oil Spills happen frequently as a result of offshore vessel failures or accidents. In this project, we present a multi-robot swarm system that will work on the water surface to help clean up the oil spill in a certain manner and also prevent it from spreading wider.

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1. Introduction:

Swarm robotics is a field of robotics research inspired by the emergent behavior of large groups of simple creatures. Swarm robotics is when many robots work together as a team to achieve a common goal. Swarm robotics has many advantages such as the ability to complete more tasks simultaneously than a traditional single based robot. Swarm robotics can use these advantages to generally perform a task faster than a single robot within the same environment.

The demand for oil is increasing every day and is substantiated by expanding submarine oil pipelines, distribution of oil and its derivatives by using tankers to carry it to many destinations. This leads to increased chances of oil leakage in the sea either by leakage from submarine oil pipelines or accidents with the tankers. In the past, such mis-happenings lead to major disasters of oil leakage in the high seas. Oil spills threaten the wildlife in the sea and hence, there is a necessity for research on cleaning up oil spills quickly and efficiently has become a very important issue by researchers and companies concerned.

The current processes to clean up oil spills take much time and are expensive. In case the water is turbulent, the spill may get spread wider making it difficult to finish the process of cleaning up. We propose a method to send a swarm of robots in the working area which can collect the oil from the water and thereby accomplish the task of cleaning the water. This methodology can provide us a cheaper, faster and efficient way to solve the problem.

This report starts with a description of the mechanical structure and various components of the robot system. It further describes the methodology of the proposed algorithm and it ends with a discussion about the problems we may face while implementing such a model.

2. Construction of Robot:

In this section, we provide information regarding the construction of each robot in the swarm. We categorize the robot into the following subsystems:

2.1) Mechanical Structure:
Compared to other robotic systems, here we need to seal the robot enclosures for protection against water. The recommended ingress protection is referred to as IP68 (or above) for all the enclosures. Failure to meet this requirement could lead to damaging of the electronics. Each robot's mechanical design comprises of the structural frame, enclosures for electronics, and actuators. Each robot has 3 degrees of freedom.

2.2) Sensors and Actuators:
Sensors: In order to complete our task, we need to sense the position, velocity, rotation and the amount of oil swept by the robot. The master has a GPS module that is used to define its position. We can determine the positions of the slaves using the XBees modules and the algorithms described later in the report. The rotation of the robot can be determined by the IMU module, which is provided to each of the robots. We use a piezoelectric sensor to sense the velocity. The amount of oil-soaked can be determined by the increasing mass of the foam.
Actuators: Actuators used by us includes servo motors and thrusters. Servo motors are used to drive the belt system which carrier the foam. We use servo motors to get better speed control. Each vehicle has 4 thrusters. Two of the thrusters provide thrust in the forward direction while the other two in the left (or right) direction. In this way, using 4 thrusters, we achieve a DOF 3 with greater stability.

2.3) Control Unit:
We use the Arduino Mega as the single microcontroller to drive all the actions of the robot. The microcontroller takes input from various sensors, communicates with other robots in swarm via communication channels and then drives the actuators accordingly. This unit is also responsible to process the algorithms for stability and driving actions of the robot.

2.4) Power Unit:
As the robot has to stay in the water bodies for a longer time, we suggest the use of solar power to provide power to all components of the robot. We also provide an additional backup power supply for the robot in the form of LiPo batteries. In case of bad weather conditions or in case of insufficient sunlight, the robot can use the power from the batteries.

2.5) Communication:
We will use the XBee modules along with a GPS module for communication. Only the master is equipped with the GPS module, while all the robots in the swarm (Master and Slaves) are equipped with the XBee modules for communication among themselves. If the robots were to communicate underwater, we cannot use the Xbee modules and GPS as the attenuation of signal strength occurs. In that case, we might need to shift to SONAR communication.

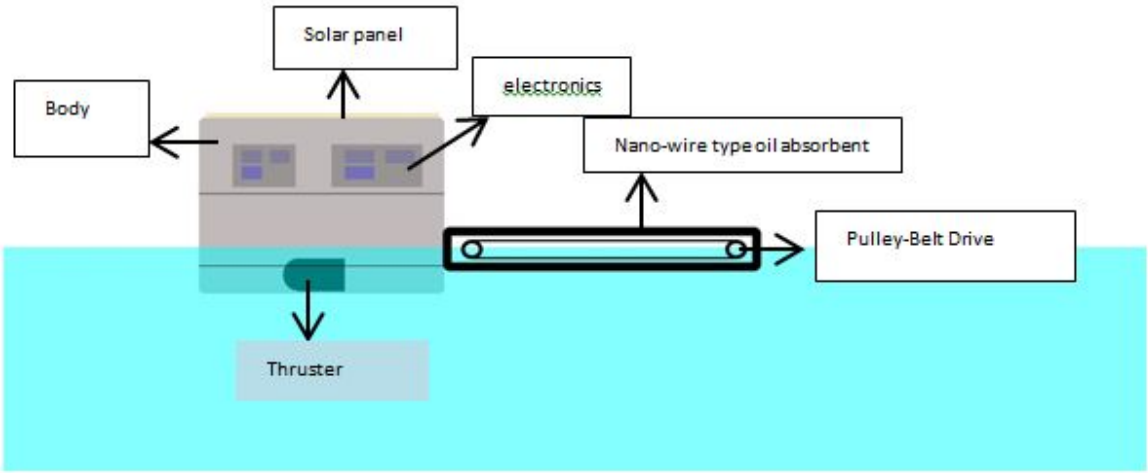

Fig 1: Schematic of a robot in the swarm.

3. Problem Statement:

Our aim is to design a swarm of robots that can clean oil spills in minimum possible time. The design of every single robot will be the same. We categorize the robots in swarm into Master and Slave. The master robot will send all the global information to all other slave robots. The communication process and controller design have to be provided. Along with this, the optimized swarm formation needs to be provided. Apart from this, it may happen that there is some defect in any one of the robots. We need to detect it and replace it as fast as possible. We also need to provide the power and cost analysis of the system.

We are taking the following assumption while providing the solution:
3.1) Oil spill has occurred in a continuous region with a single boundary, this means that the oil spill region is continuous and not discrete.
3.2) The climate when cleaning takes place is quite clear i.e sun is shining brightly and there is no chance of storms.

Foam in this report refers to an Oil-absorbing nano-wire mesh material developed at MIT. The material can absorb up to 20 times its weight in oil. The paper is recyclable, allowing it to be a reusable resource for future oil spills.

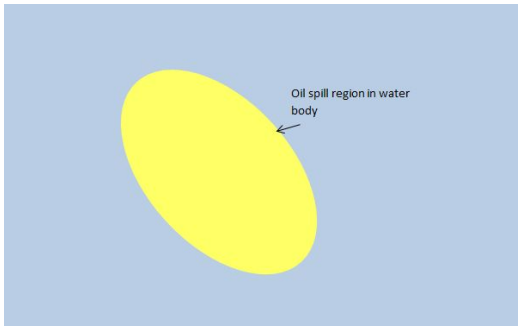

Fig.2: Represents the assumed continuous model of oil spill

3. Communication:

Communication is one of the most important aspects of Swarm Robotics. The robots are required to communicate their current parameters such as their positions and velocities to the surrounding robots in the swarm in order to perform the task in an efficient manner. The communication modules we use include the GPS module and the XBee module. The Master robot is fed with the location of the oil spill and it uses the GPS to travel to the location while the Slave robots follow the Master in a particular formation. The XBee module is used by the swarm to communicate among themselves. An XBee is a wireless transceiver used for bidirectional communication at moderate speeds. The XBee is connected to an XBee Explorer board which is used as an interface between a computer/microcontroller (Arduino MEGA in our case) and the XBee. The XBee module can be used in the following two modes:

1. Peer to Peer (P2P) Communication: This mode is used for communication between two robots. Two XBees can communicate with each other, only if their Personal Area Network ID (PAN ID) is the same. The PAN ID is a hexadecimal number ranging between 0x0 and 0xffff. Since there are 65536 possibilities, there are very fewer chances of faulty/undesirable communication. Each XBee must be assigned a source address which is called MY address and a destination address. Both the addresses are between 0x0000 and 0xFFFF. For one XBee to be able to send data to another, it must have the same destination address as the other XBees source. It may happen that the communication between two XBees is one-way or both-ways depending upon the source address and the destination address.
2. Broadcast Mode Communication: This mode is used by a robot in a swarm to broadcast a message to the robots connected to a particular network. XBees can share the data with other XBees in the same network. The broadcast mode communication using XBee requires one Coordinator and other End Devices.

In our case, as the robots are above the surface of the water, we can use the XBee modules along with a GPS module for communication. If the robots were to communicate underwater, we cannot use the Xbee modules and GPS as the attenuation of signal strength occurs.

Another common form of wireless communication that we can use underwater is sonar or acoustic communication. This type of communication works on the same principles as a standard transceiver but operates at a much lower frequency. It is well known that this type of method can be used to communicate underwater as many underwater mammals, such as whales, use these types of frequencies to communicate with each other over significantly large distances. The main drawback of using this type of communication is that a relatively low bandwidth is achieved and the signal travels relatively slowly.

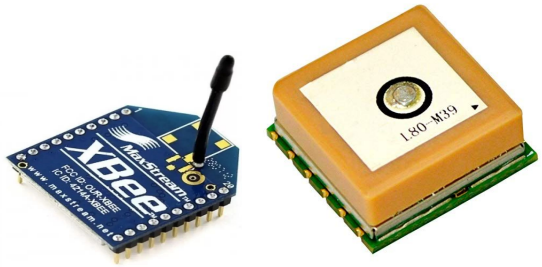

Fig.3: Communication Modules: XBee (left), GPS (right)

4. Working:

We provide the master robot with the location of the oil spill. The robots are deployed. The master robot has a GPS module which assists the Master to go to the location. The master robot and the slaves communicate with each other using the XBee modules. The robots move as a swarm until they reach the target location. After they reach the target location the oil is detected by the foam on the belt drive. This then provides the feedback to the servo motors which controls the speed for the belt drive. The foam on the belt can soak up to 100 times its weight in oil. The robots move in a position such that they clean up the oil spill while also ensuring that the spill doesn't spread. The power for all the activities is provided by solar energy. The algorithms described in the report provide the microcontroller of the data for the motion of the robot. This then is provided to the thrusters which actuate the motion. Also, the sensors ensure that the error margin is the minimum.

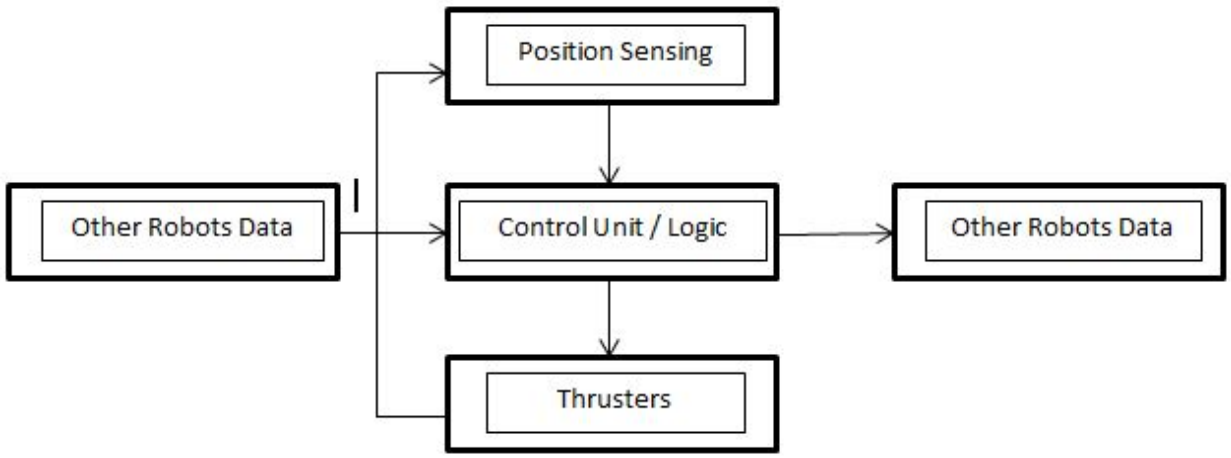

Fig. 4: Block Diagram for Motion of a Robot in a swarm (Closed Loop).

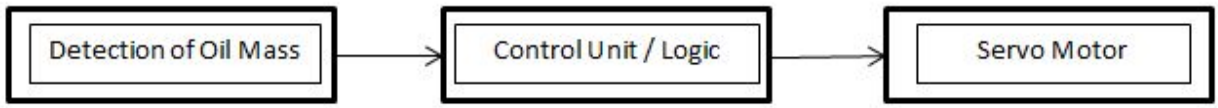

Fig .5: Block Diagram for Servo Motor Actuation (Open Loop)

5. Swarm Formation:

This is the task where all the robots need a proper algorithm to work together in a synchronized manner. Before the application of the algorithm, the robots need all the global information and local information to feed to the algorithm. We have various sensors for gathering local information such as distance from the neighboring robot, oil concentration, etc. For global information such as the center of the oil spill, the boundary of the oil spill, etc. are gathered by the GPS available on the master robot. This information is then communicated to all other robots by sequential broadcasting to neighboring robots.

5.1) Maintaining a dynamic graph structure:

The master robot moves to the center of the oil spill. The center can move from its position and when it happens, the master robot needs to shift its position to the new center. Other robots get the information about the centre by this master robot which broadcasts the message to all other neighboring robots, all of which transfers further. The whole network of robots can be then constructed into a graph network with edge weights equal to the distance between the robots:

1. Each robot sends three main information packets to all other neighboring robots-
   - position of the master robot and a reference axis,
   - their own position with respect to the master robot,
   - position of each and every receiver with respect to itself.

   Using these three positions, every bot can get their own location and also send out the necessary information to their neighboring robots.

2. Knowing the above information, every robot can build two types of graphs. First, a graph with nodes containing information about the neighboring robots. This graph can be made quickly and efficiently. It is used to avoid collision and to maintain a proper speed. Second, a graph with nodes containing information about all the robots in the swarm. This graph helps the robot to find the way back to the home ship in the minimum time possible.

5.2) Detecting the edge of the oil spill:
When the robots enter the region of the oil spill, the nanowire mesh paper starts absorbing oil. This results in a change in the weight of the paper. This change, if found to be greater than a certain calibrated threshold value, can detect the boundaries of the oil spill region.

5.3) Lining of robots on the edge:
Robots need to be lined up on the edge of the oil spill so as to prevent further expansion of the spill area. They are able to do this by the following algorithm:

1. A certain number of robots are assigned the task of lining up on the circumference. This number is calculated by the following formula:

$$n = (circumference/area) \times total\ number\ of\ robots$$

2. These n robots need to constantly detect the edge and need to be on the edge. Moreover, they also need to move along the edge continuously so as to prevent a further increase in the oil spill region.
3. As the circumference changes the ratio is again calculated and either extra robots are made to join this category or robots already in this category are made to stop lining up.

5.4) Clustering of robots in the area of the oil spill:
Apart from the robots which are lining up on the edge, there remain a lot more robots. These are sent in the area of oil spill towards the center.

1. The Xbee sensor provided on each robot helps them to maintain a minimum distance between neighboring robots. The motion is illustrated in the figure below.
2. If the robot has to return back to its home/ship (when the nanowire paper no longer absorbs oil), the path to home is created by finding the shortest path from the graph created.

5.5) Benefits of such formation:
This kind of formation would have the following benefits:

1. The area of an oil spill would not increase.
2. All the robots work simultaneously until the task ends.
3. In case of failure, it is easier for a robot to find its way back.
4. Information from the master robot can be broadcasted to all the robots very quickly, therefore in case of any unwanted climate change all the robots can be instructed to go back to the ship

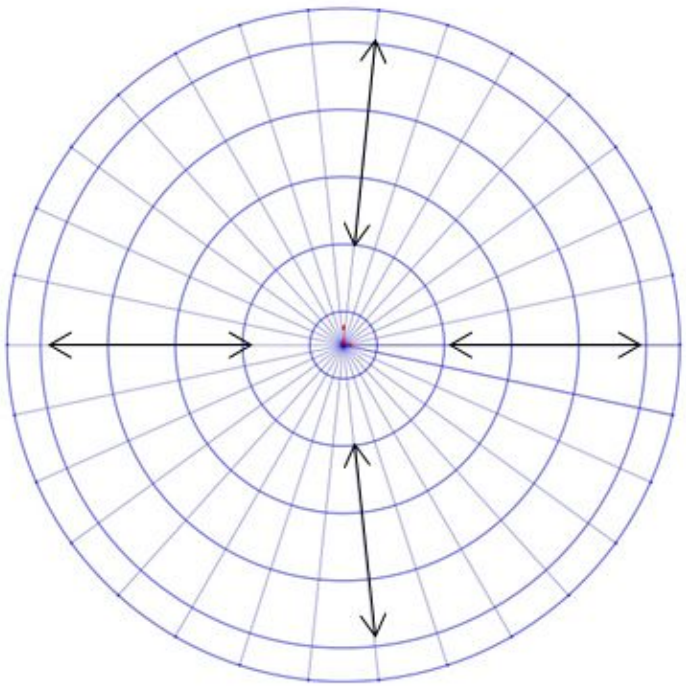

Fig 6: All the intersection points represents the robots in the swarm. The robots on the outermost boundary do not move radially unless there is a decrease in the radius of the oil spill. All the inner robots move to and fro radially. All the robots move along the circle.

6. Future work:

We have solved the problem in case there is a continuous oil spill. But we need to design different formation in case of varying oil spill structures. They can be intermittent, in this case, we would need a completely different formation for cleaning up the sea.
Moreover, we can also design an algorithm that would choose an optimal number of robots required to do the cleaning operation.

7. References:

1. All figures in this report (except fig 3) were made using Modelling Softwares.
2. MIT develops a 'paper towel' for oil spills.[Link]
3. Oil Spill trajectory tracking using Swarm Intelligence and Hybrid fuzzy system [Link].
4. Analysis of Prediction Models for Swarm Robotics [Link]